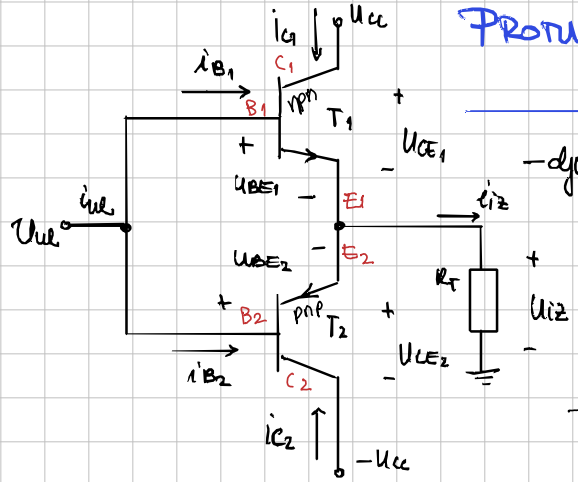
PROTUTAKTNO POJAČALO SNAGE KLASE B



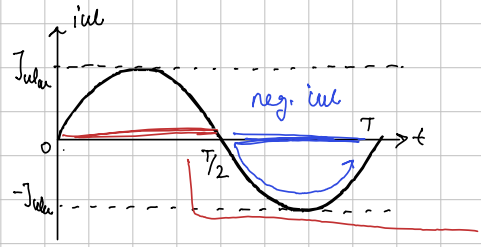
- djelotvornost se povećava uklanjanjem istosmjerne snage trošila

→ bude na usmenom!

→ želimo izbjeći DC struju na trošilu

↳ tranzistori se ne postavljaju u SRT

Opis rada

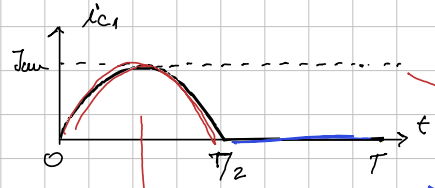


$i_{ul} = i_{B1} + i_{B2}$

propusno polarizirano pn spoj BE npn T_1

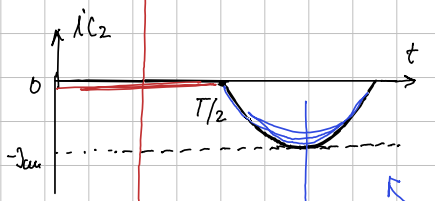
zaporno polarizira pn spoj BE pnp T_2

$i_{ul} = i_{B1}$, $i_{B2} = 0$



$i_{C1} = \beta_1 \cdot i_{B1}$, $i_{C2} = 0$

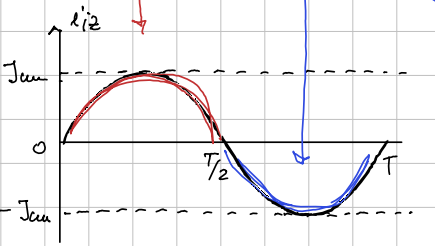
$i_{ul} > 0$
vodi T_1
ne vodi T_2



- zaporno pn spoj od npn T_1
- propusno pn spoj od pnp T_2

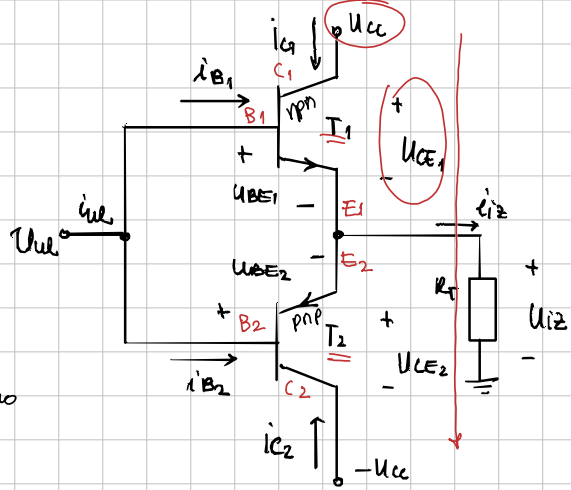
$i_{C1} = 0$ $i_{C2} = \beta_1 \cdot i_{B2}$

$i_{ul} < 0$
ne vodi T_1
vodi T_2



$u_{iz} = i_{iz} R_T = (i_{C1} + i_{C2}) R_T$

Radne staze tranzistora



$$u_{iz} = u_{CC} - u_{CE1} = -u_{CC} - u_{CE2}$$

$$u_{CE1} = 2u_{CC} + u_{CE2}$$

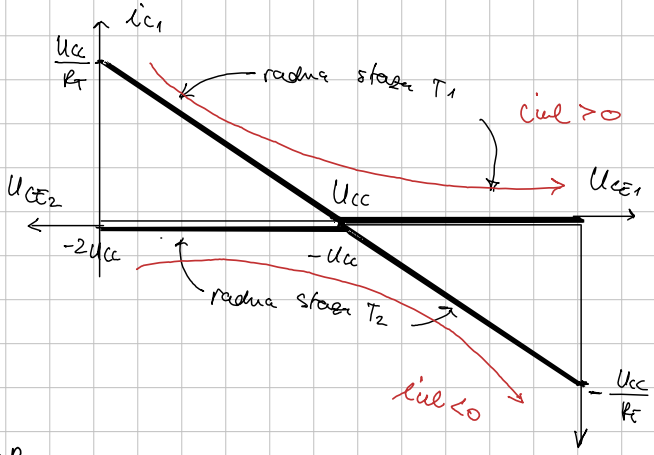
$$-u_{CE2} = 2u_{CC} - u_{CE1}$$

→ izlazni naponi tranzistora međusobno su ovisni

Za $i_{ul} = 0$ → tranz ne vode

$u_{iz} = 0$

$u_{CE1} = u_{CE2} = u_{CC}$



Za $i_{ul} > 0$ vodi T_1

$$u_{CE1} = u_{CC} - i_{iz} \cdot R_T = u_{CC} - i_{C1} \cdot R_T$$

Za $i_{ul} < 0$ vodi T_2

$$-u_{CE2} = u_{CC} + i_{iz} R_T = u_{CC} + i_{C2} \cdot R_T$$

Odnosi snaga

Snaga izvora napajanja

$$p_{CCP}(t) = \begin{cases} u_{CC}\, i_{C1} & \text{za } 0 < t < T/2 \\ 0 & \text{za } T/2 < t < T \end{cases} \qquad p_{CCN}(t) = \begin{cases} 0 & \text{za } 0 < t < \frac{T}{2} \\ -u_{CC}\, i_{C2} & \text{za } T/2 < t < T \end{cases}$$

$$P_{CCP} = \frac{1}{T}\int_0^T p_{CCP}(t)\,dt = \frac{1}{2\pi}\left[\int_0^{2\pi} U_{CC} I_{am} \cdot \sin(\omega t)\,dt + \int_{\pi}^{2\pi} 0\,dt\right] = \frac{U_{CC} I_{am}}{\pi}$$

$$P_{CCN} = \frac{1}{T}\int_0^T p_{CCN}(t)\,dt = \frac{1}{2\pi}\left[\int_0^{\pi} 0\,dt - \int_{\pi}^{2\pi} U_{CC} I_{am} \sin(\omega t)\,dt\right] = \frac{U_{CC} I_{am}}{\pi}$$

$$\boxed{P_{CC} = P_{CCP} + P_{CCN} = \frac{2}{\pi} U_{CC} I_{am}}$$

Snaga trošila

$$p_{RT}(t) = R_T \cdot i_{iz}^2 = R_T (i_{c_1} + i_{c_2})^2 \qquad i_{iz} = J_{cm} \sin \omega t$$

$$P_{RT} = \frac{1}{T}\int_0^T p_{RT}(t)\,dt = \frac{1}{2\pi}\int_0^{2\pi} R_T J_{cm}^2 \sin^2(\omega t)\,dt = R_T \frac{J_{cm}^2}{2}$$

javlja samo izmjenična snaga!

Snage tranzistora

$$2p_T(t) = p_{CC}(t) - p_{RT}(t)$$

potrošnju snage Tranz. moramo ograničiti zbog zagrijavanja

$$p_{T_1}(t_1) = \begin{cases} u_{CE}\, i_{c_1} - R_T i_{c_1}^2, & 0 < t < \frac{T}{2} \\ 0, & T/2 < t < T \end{cases}$$

$$P_{T_2}(t) = \begin{cases} 0, & 0 < t < T/2 \\ -u_{CC} \cdot i_{c_2} - R_T \cdot i_{c_2}^2, & T/2 < t < T \end{cases}$$

Temp se mijenja znatno sporije od signala → kritična je srednja SNAGA:

$$2P_T = P_{CC} - P_{RT} \qquad \frac{2}{\pi} U_{CC} \cdot J_{cm}, \quad P_{RT} = R_T \frac{J_{cm}^2}{2}$$

$$P_T = \frac{P_{CC} - P_{RT}}{2} = U_{CC} \cdot \frac{J_{cm}}{\pi} - R_T \frac{J_{cm}^2}{4}$$

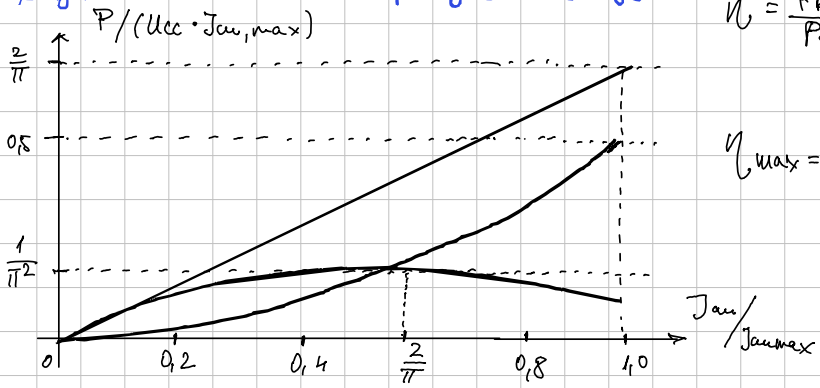
Maksimalna snaga → koju maksimalnu snagu tranzistori mogu podnijeti?

$$\frac{\partial P_T}{\partial J_{cm}} = \frac{U_{CC}}{\pi} - R_T \frac{J_{cm}}{2} = 0 \longrightarrow J_{cm} = \frac{2}{\pi}\frac{U_{CC}}{R_T}$$

ekstrem

$$P_{T,max} = \frac{U_{CC}^2}{\pi^2 R_T}$$

Djelotvornost i raspodjela snaga



$$\eta = \frac{P_{RT}}{P_{CC}} = \frac{\pi}{4}\frac{R_T \cdot J_{cm}}{U_{CC}}$$

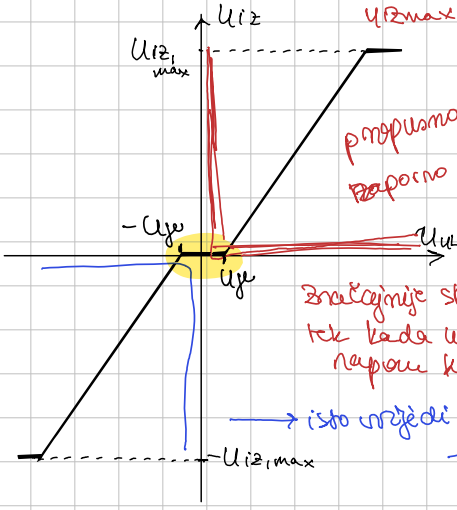
$$\eta_{max} = \frac{\pi}{4}\frac{R_T \cdot J_{cm,max}}{U_{CC}} = \frac{\pi}{4} = 78{,}5\%$$

$$\frac{U_{CC}}{R_T}$$

značajno pojačanje u odnosu na tip A sa 25%

Prijenosna karakteristika – preskočna izobličenja

$u_{izmax} = U_{CC} - U_{CE,zas}$

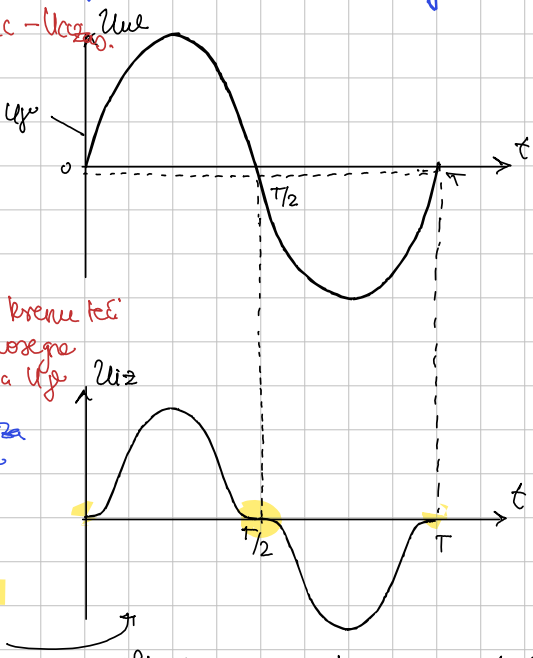


propusno T1

zaporno T2

značajnije struje kreću teći tek kada u_{ul} dosegne napon kolena U_γ

→ isto vrijedi ali za $-U_\gamma$



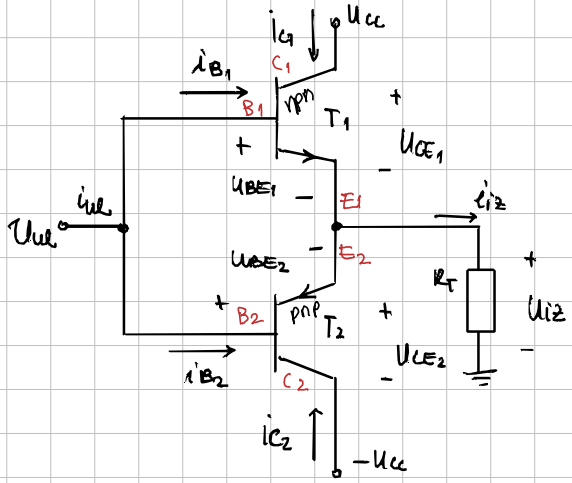
Za $|u_{ul}| < U_\gamma$ → mrtvo područje

posljedica je preskočno izobličenje

$u_{izmax} \approx U_{CC}$

$-u_{izmax} \approx -U_{CC}$

$u_{izm} \approx U_{CC} \rightarrow J_{izm} \approx \frac{U_{CC}}{R_T}$



Primjer 4.2)

Pretvarač = 20W

$R_T = 8\,\Omega$

$u_{izmax} = ?$ $\quad u_{izmax} = 1V + u_{izmax}$ (U_{CC})

$u_{iz} = R_T \cdot i_{iz}$

$$P_{R_T} = \frac{R_T \cdot J_{izm}^2}{2} = \frac{R_T \cdot J_{iz}^2{}_m}{2} = \frac{u_{izm}^2}{2 \cdot R_T}$$

$$u_{iz\,max} = \sqrt{2 R_T \cdot P_{R_T}} = \sqrt{2 \cdot 8\Omega \cdot 20W} \rightarrow u_{izmax} = 17.9V$$

$\rightarrow \boxed{U_{CC} = 18.9V}$

$$J_{iz\,max} = J_{izm} = \frac{u_{izm}}{R_T} = \frac{17.9}{8} = 2.24\,A$$

$$P_{CC} = \frac{2}{\pi} U_{CC} \cdot J_{izm} = 27.1\,W$$

$$P_{izmax} = \frac{U_{CC}^2}{\pi^2 R_T} = 4.57\,W$$

$$\eta = \frac{P_{R_T}}{P_{CC}} = 0{,}738 = 73.8\% < 78.5\%$$